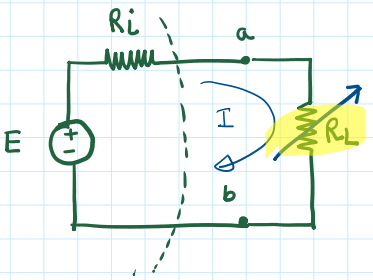
Lecture 6: Thevenin, Norton, Maximum Power Transfer and Star-Delta Transformation

Maximum Power Transfer Theorem :

A maximum power has to be transferred to the load. What will be the value of R_L ?



$$P_L = I^2 * R_L \;;\; I = \frac{E}{R_i + R_L}$$

$$= \left(\frac{E}{R_i+R_L}\right)^2 * R_L = \frac{E^2}{(R_i+R_L)^2} * R_L$$

$$\frac{d}{dx}(uv) = \frac{u'v - uv'}{v^2}$$

$$\frac{dP_L}{dR_L} = E^2 \left[\frac{1\cdot(R_i+R_L)^2 - R_L\, 2(R_i+R_L)\cdot 1}{(R_i+R_L)^4}\right] = 0$$

$$\Rightarrow (R_i+R_L)^2 = 2R_L(R_i+R_L)$$

$$\boxed{R_L = R_i}$$

$$\left.\frac{d^2P_L}{dR_L^2}\right|_{R_L=R_i} = -\text{ve Term. (verify)} = \frac{d}{dR_L}\, E^2 \frac{(R_i+R_L)(R_i+R_L-2R_L)}{(R_i+R_L)^4}$$

$$= E^2 \frac{d}{dR_L}\left\{\frac{(R_i^2-R_L^2)}{(R_i+R_L)^4}\right\}$$

$$= E^2 * \frac{-2R_L(R_i+R_L)^4 - (R_i-R_L^2)\,4(R_i+R_L)^3\cdot 1}{(R_i+R_L)^8}$$

$$= E^2 \frac{(R_i+R_L)^3\{-2R_L - 4(R_i-R_L)\}}{(R_i+R_L)^8}$$

$$= E^2 \left\{\frac{-2R_L}{(2R_L)^4}\right\} \quad \text{At } R_L = R_i$$

$$= -\frac{E^2}{(2R_L)^3} = \text{Negative value.}$$

The value of maximum power is given as,

$$P_{Lmax} = \frac{E^2}{(R_L+R_L)^2} * R_L = \frac{E^2}{4R_L} \quad \ldots (\$)$$

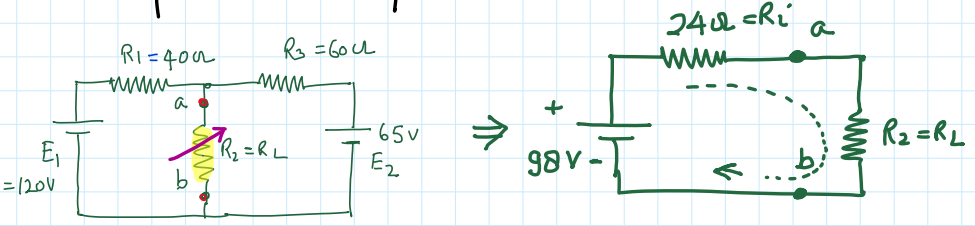
$$P_{Lmax} = \frac{E^2}{(R_i+R_L)^2} \cdot R_L = \frac{E^2}{4R_L} \quad \text{---(\#)}$$

$$P_{in} = E * \frac{E}{(R_i+R_L)} = \frac{E^2}{2R_L} \quad \text{---(\#)}$$

$$\eta = \frac{P_{Lmax}}{P_{in}} = \frac{E^2/4R_L}{E^2/2R_L} = 0.5 = 50\%$$

Maximum power transfer theorem refers to impedance matching
i.e. $R_L = R_i$

Consider the previous circuit, that we considered for Thevenin's equivalent circuit.



For maximum power in R_L = $R_L = R_i = 24\Omega$

$$I = \frac{98}{24+24} = 2.041\ A$$

$$P_L = P_{max} = I^2 \times 24 = 100.04\ W$$

$$P_{in} = I^2 * (24+24) = 200.08\ W$$

$$\eta_{max} = \frac{P_{max}}{P_{in}} = \frac{100.04}{200.08} = 0.5 \text{ or } 50\%$$

Applications of Maximum Power Transfer Theorem:

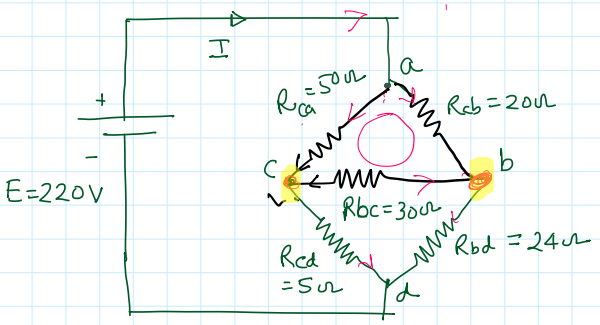
1. Communication system; where maximum power transfer is important for efficient signal transmission
2. Audio Engineering: amplifier's output impedance matches the speaker impedance.
3. Wireless Power Transfer: Source and receiver circuits are tuned for resonant coupling to maximize power flow.

Star-Delta Transformation:

Star-Delta Transformation:

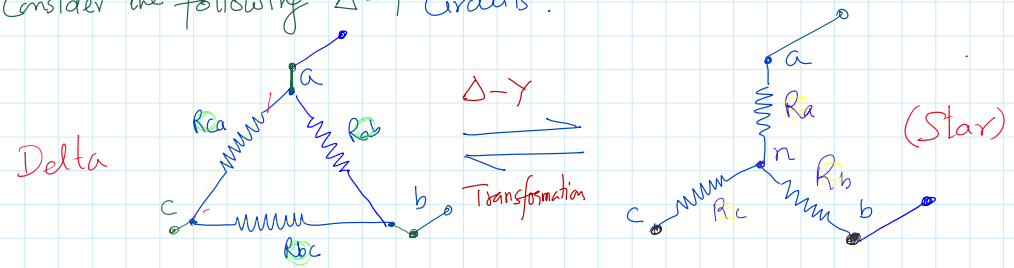
Example 7:

Consider the following circuit



* These resistances are neither in series nor in shunt. They are actually in form of Δ or Y connection. How to solve such circuits?

Consider the following Δ-Y Circuits:



Thus, what is the relationship between R_{ab}, R_{bc}, R_{ca} (Δ-connected) and R_a, R_b R_c (Y connected)

We shall establish this relationship.

Star Side Delta Side

$$R_a+R_b = \frac{R_{ab}*(R_{bc}+R_{ca})}{R_{ab}+R_{bc}+R_{ca}} \quad ----(1)$$

$$R_b+R_c = \frac{R_{bc}(R_{ca}+R_{ab})}{R_{ab}+R_{bc}+R_{ca}} \quad ----(2)$$

$$R_c+R_a = \frac{R_{ca}(R_{ab}+R_{bc})}{R_{ab}+R_{bc}+R_{ca}} \quad ----(3)$$

Adding ①-③

$$(R_a+R_b+R_c) = \frac{R_{ab}(R_{bc}+R_{ca})+R_{bc}(R_{ca}+R_{ab})+R_{ca}(R_{ab}+R_{bc})}{2(R_{ab}+R_{bc}+R_{ca})} \quad -④$$

$$-(R_b+R_c) = \frac{-2R_{bc}(R_{ca}+R_{ab})}{2(R_{ab}+R_{bc}+R_{ca})}$$

$$R_a = \frac{R_{ab}(R_{bc}+R_{ca})-R_{bc}(R_{ca}+R_{ab})+R_{ca}(R_{ab}+R_{bc})}{2(R_{ab}+R_{bc}+R_{ca})}$$

$$= \frac{R_{ab}R_{bc}+R_{ca}R_{ab}-R_{bc}R_{ca}-R_{ab}R_{bc}+R_{ca}R_{ab}+R_{bc}R_{ca}}{2(R_{ab}+R_{bc}+R_{ca})}$$

$$= \frac{R_{ab}R_{bc} + R_{ca}R_{ab} - R_{bc}R_{ca} - R_{ab}R_{bc} + R_{ca}R_{ab} + R_{bc}R_{ca}}{2(R_{ab}+R_{bc}+R_{ca})}$$

$$= \frac{2\,R_{ca}R_{ab}}{2\,\Sigma R_{ab}} = \frac{R_{ab} * R_{ca}}{\Sigma R_{ab}}$$

Thus we have

$$R_a = \frac{R_{cb} * R_{ca}}{\Sigma R_{ab}}$$

$$R_b = \frac{R_{bc} * R_{ab}}{\Sigma R_{ab}}$$

$$R_c = \frac{R_{ca} * R_{bc}}{\Sigma R_{ab}}$$

$\Delta - Y$ Transformation

-------- (*)

Now let us consider $Y \to \Delta$ Transformation:

Given: R_a, R_b, R_c compute R_{ab}, R_{bc}, R_{ca}.

From (*)

$$\frac{R_a}{R_b} = \frac{R_{ab} * R_{ca}}{R_{bc} * R_{ab}} = \frac{R_{ca}}{R_{bc}} \quad \cdots\cdots (1)'$$

Similarly $\frac{R_b}{R_c} = \frac{R_{ab}}{R_{ca}}$ ------ (2)' $\Rightarrow R_{ca} = \frac{R_c}{R_b} * R_{ab}$

$\frac{R_c}{R_a} = \frac{R_{bc}}{R_{ab}}$ ------ (3)' $\Rightarrow R_{bc} = \frac{R_c}{R_a} * R_{ab}$

Now take first equation:

$$R_a + R_b = \frac{R_{ab}(R_{bc}+R_{ca})}{R_{cb}+R_{bc}+R_{ca}} = \frac{R_{ab} * \left(\frac{R_c}{R_a} R_{ab} + \frac{R_c}{R_b} R_{ab}\right)}{R_{ab} + \frac{R_c}{R_a} R_{ab} + \frac{R_c}{R_b} R_{ab}}$$

$$(R_a + R_b) = \frac{\frac{R_{ab} R_c (R_a+R_b)}{(R_a R_b)}}{\frac{R_a R_b + R_b R_c + R_c R_a}{R_a R_b}}$$

$\Rightarrow$

$$R_{ab} = \frac{R_a R_b + R_b R_c + R_c R_a}{R_c} = \frac{\Sigma R_a R_b}{R_c}$$

Similarly

$$R_{bc} = \frac{R_a R_b + R_b R_c + R_c R_a}{R_a} = \frac{\Sigma R_a R_b}{R_a}$$

$$R_{ca} = \frac{R_a R_b + R_b R_c + R_c R_a}{R_b} = \frac{\Sigma R_a R_b}{R_b}$$

$Y - \Delta$ Transformation

Now consider this problem with star-delta transformation.

Example 8: Consider the following circuit

Now consider this problem with Star - delta transformation.

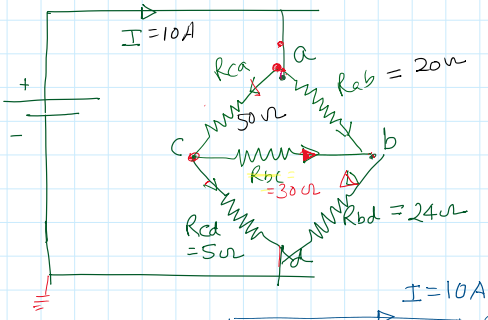
Example 8 : Consider the following circuit

Determine currents in all resistors shown in the circuit.

Mesh Method : $m = b - (j-1)$

$= 6 - (4-1) = 3$

Nodal Method $j - 1 = 4 - 1 = 3$

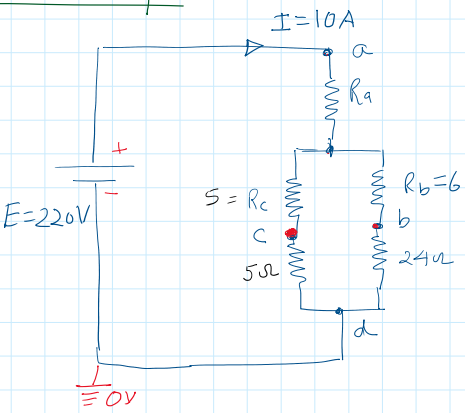


* These resistances are neither in series nor in shunt. They are actually in form of Δ or Y connection. How to solve such circuits?

$$R_a = \frac{R_{ab} \times R_{ca}}{R_{ab}+R_{bc}+R_{ca}} = \frac{50 \times 20}{50+20+30} = 10\,\Omega$$

$$R_b = \frac{R_{bc} * R_{ab}}{\sum R_{ab}} = \frac{20 \times 30}{100} = 6\,\Omega$$

$$R_c = \frac{R_{ca} \times R_{bc}}{\sum R_{ab}} = \frac{30 \times 50}{100} = 15\,\Omega$$

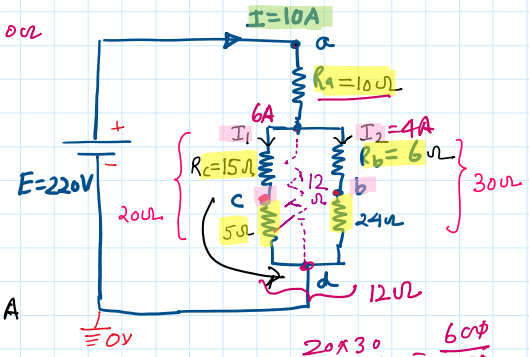


$$R_a = \frac{R_{ab} * R_{ca}}{R_{ab}+R_{bc}+R_{ca}} = \frac{20 \times 50}{20+30+50} = \frac{1000}{100} = 10\,\Omega$$

$= 10\,\Omega$

$$R_b = \frac{R_{bc} \times R_{ab}}{\sum R_{ab}} = \frac{20 \times 30}{100} = 6\,\Omega$$

$$R_c = \frac{R_{ca} * R_{bc}}{\sum R_{ab}} = \frac{30 \times 50}{100} = 15\,\Omega$$



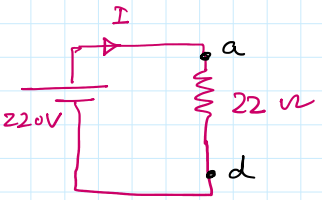
$$I = \frac{E}{R_T} = \frac{E}{R_{ad}} = \frac{220}{22} = 10\,A$$

$$\frac{20 \times 30}{20+30} = \frac{600}{50} = 12$$

$$I_1 = I * \frac{30}{20+30} = 6\,A$$

$$I_2 = I * \frac{20}{20+30} = 4\,A$$

$$I_1 + I_2 = I = 10\,A$$



$$I_{cd} = I_{5\Omega} = 6\,A$$
$$I_{bd} = I_{24\Omega} = 4\,A$$

$$V_{cb} = V_c - V_b = V_{cd} - V_{bd}$$

$$V_{cd} = I_{cd} * 5 = 6 \times 5 = 30V, \quad V_{bd} = 4 \times 24 = 96V$$

$V_{cd} = I_{cd} \times 5 = 6 \times 5 = 30V$, $V_{bd} = 4 \times 24 = 96V$

$V_{cb} = V_{cd} - V_{bd} = 30 - 96 = -66\,V$

$I_{cb} = \frac{V_{cb}}{30} = \frac{-66}{30} = -2.2A$

$I_{bc} = -I_{cb} = 2.2A$

$I_{ac} = \frac{V_a - V_c}{R_{ca}} = \frac{220 - 30}{50} = 3.8A$

$I_{ab} = \frac{V_a - V_b}{R_{ab}} = \frac{220 - 96}{20} = 6.2A$

Once, these voltages are known, we can compute currents in the original circuit; i.e., I_{ab}, I_{ac}.

$I_{cb} = \frac{V_{cd} - V_{bd}}{30} = \frac{-66}{30} = -2.2A$

or $I_{bc} = 2.2A$

$I_{ab} = \frac{V_a - V_b}{20} = \frac{V_{ad} - V_{bd}}{20}$

$= \frac{220 - 96}{20} = \frac{124}{20} = 6.2A$

$I_{ac} = \frac{V_{ad} - V_{cd}}{50} = \frac{220 - 30}{50} = \frac{190}{50}$

$= 3.8A$

